

RECOMMENDATION SYSTEMS FOR PERSONALIZED CONTENT DISCOVERY

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PROBLEM OVERVIEW

THE BURNING PLATFORM

Netflix Prize Dataset

100M+ RATINGS

480K USERS

17.7K MOVIES

HIGHLY SPARSE

Objective

Build a recommendation system capable of:

- Learning user preferences
- Predicting unseen ratings
- Generating personalized recommendations
- Improving content discovery

Models Implemented

- SVD (Primary Model)
- User-Based KNN
- Item-Based KNN

DATASET & PREPROCESSING

METRIC	SAMPED DATASET
RATINGS	9.4 MILLION
USERS	96650
ACTIVE USERS	85337
MOVIES	17765
RATING SCALE	1-5

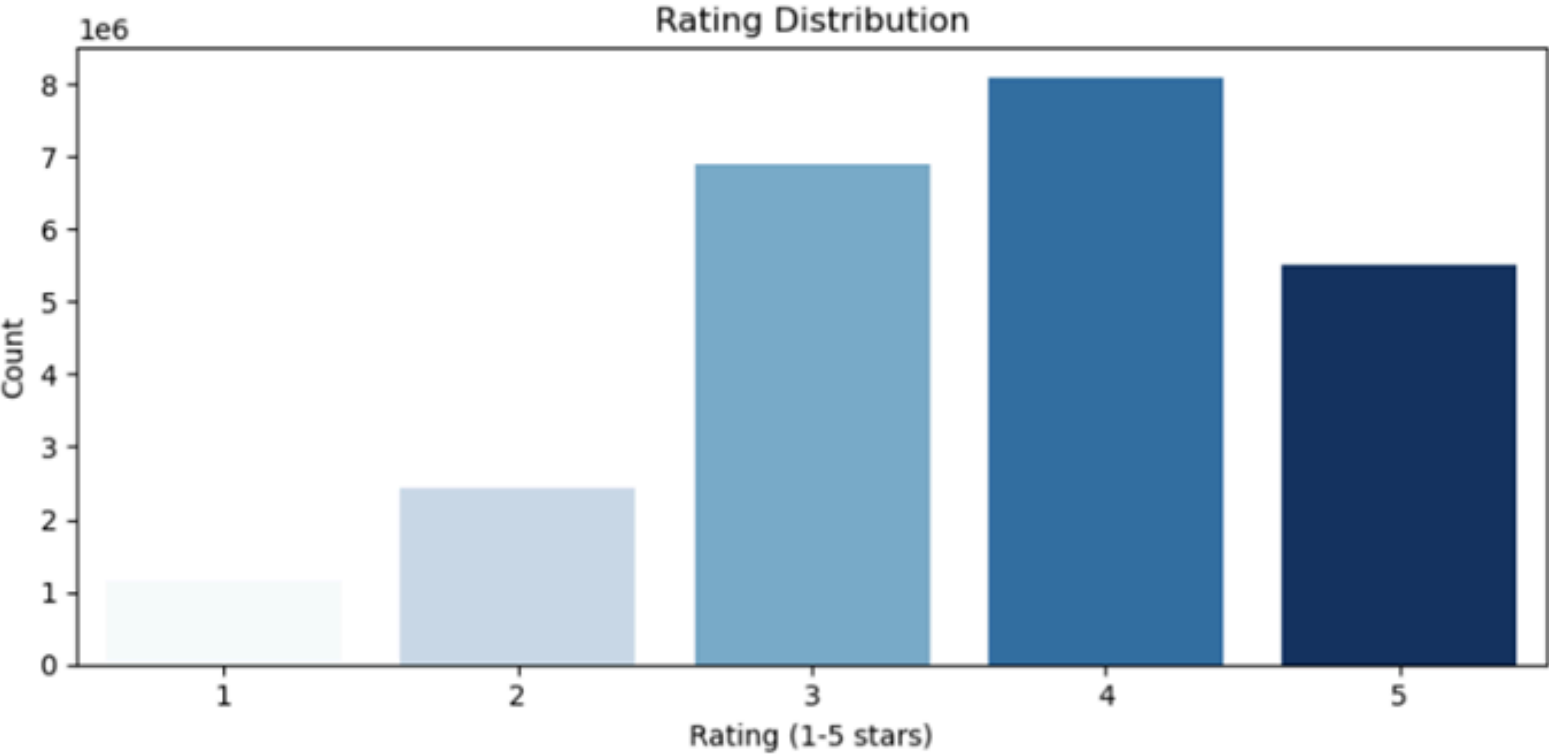
Preprocessing

- Sampled 1 in every 10 users
- Removed users with <20 ratings
- Time-based 80/20 train-test split
- Sparse matrix representation using CSR format

Why?

- Reduced memory requirements
- Removed extreme cold-start users
- Prevented temporal data leakage

EXPLORATORY DATA ANALYSIS



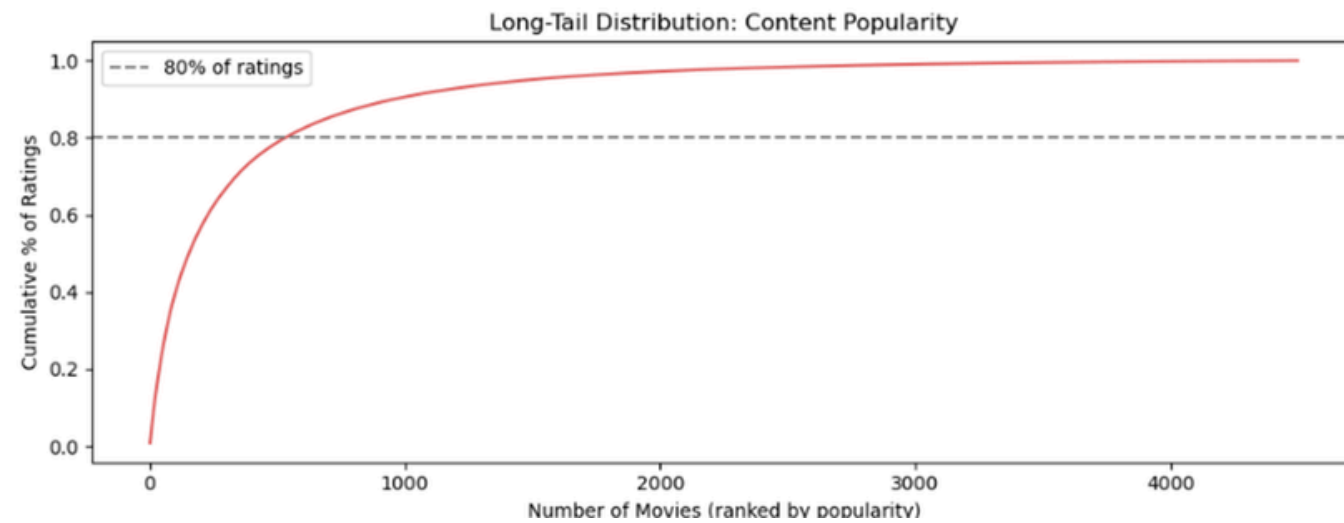
User Behaviour

- 88.9% users average above 3 stars
- 10,862 users only give 5-star ratings

SEGMENT	AVG RATING
CASUAL	3.7
MODERATE	3.65
POWER USERS	3.23

Insight: More active users rate more critically.

EDA INSIGHTS



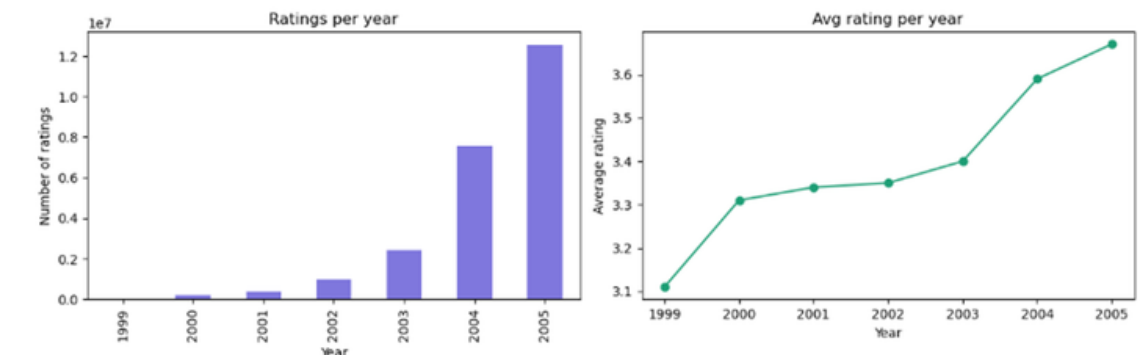
Long Tail Effect

- Top 530 movies generate 80% of all ratings
- Most-rated movie received 193,941 ratings



Data Sparsity

- 98.86% missing interactions
- Major challenge for collaborative filtering



Temporal Trends

- Ratings increased from 426 (1999)
- To 12.5M+ (2005)

RECOMMENDATION MODELS

Model 1: SVD (Primary)

- Matrix Factorization:
- $R \approx \mu + b_u + b_i + PQ^T R \approx \mu + b_u + b_i + P Q^T$
- 50 latent factors
- SGD training
- 20 epochs

1

Model 2: User-Based KNN

- $k = 20$ neighbours
- Cosine similarity
- User-user preference matching

2

Model 3: Item-Based KNN

- $k = 20$ neighbours
- Item similarity matrix
- Recommendations from similar movies

3



EVALUATION & RESULTS

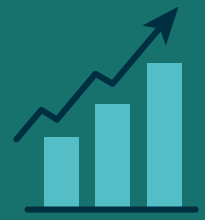
Evaluation Metrics

- **RMSE**
- **MAP@10**

Winner

✓ SVD achieved best performance on both metrics.

MODEL	RSME	MAP@10
SVD (Matrix factorisation)	0.9305	0.0235
USER-KNN	0.9957	0.0217
ITEM-KNN	1.0240	0.0111



RECOMMENDATION ANALYSIS

Success Case

User 698451

Previously liked:

- Gladiator
- LOTR
- Star Wars
- GoodFellas

Recommended:

- LOTR: Two Towers
- The Godfather
- Raiders of the Lost Ark
- Saving Private Ryan

**4/10 recommendations matched
future relevant movies**

Failure Cases

Users with:

- <25 training ratings
- Sparse interaction history
- Niche preferences

Challenges:

- Cold start
- Popularity bias
- Preference drift over time

CONCLUSIONS

Conclusions

- Compared three recommendation approaches
- SVD outperformed collaborative filtering baselines
- Successfully generated personalized recommendations
- Demonstrated recommendation quality using RMSE and MAP@10

Future Improvements

- Neural Collaborative Filtering
- Hybrid Recommendation Systems
- Larger candidate pools
- Explainable Recommendations
- Interactive Dashboard / Deployment

Key Takeaway

Latent-factor models (SVD) are more effective than neighbourhood-based methods on large, sparse recommendation datasets.